

Topic P1

Motion, forces and energy

Concept guide · Worked examples · Exam-style practice · Fully explained answers

What this guide teaches

How to measure and describe motion; connect forces to acceleration and momentum; distinguish mass, weight, density and pressure; analyse moments and stability; and calculate energy, work, power and efficiency.

Syllabus map

P1.1	Physical quantities and measurement
P1.2	Motion
P1.3	Mass and weight
P1.4	Density
P1.5	Forces, moments, centre of gravity
P1.6	Energy, work and power
P1.7	Pressure

How to study: retrieve each definition from memory, cover the worked example, then complete the practice without looking at the solution.

1. Measurement, scalars and vectors

A physical quantity is a measurable property written as a numerical value and a unit. A measurement is only useful when both are stated. Use SI units: metre (m), kilogram (kg), second (s), newton (N), joule (J), watt (W) and pascal (Pa). Convert units before substituting into an equation.

Scalar quantities have magnitude only, for example distance, speed, mass, time and energy. Vector quantities have magnitude and direction, for example displacement, velocity, acceleration, force and momentum. Two vectors in opposite directions must be treated with opposite signs.

Good measurement practice

- Read a scale at eye level to avoid parallax error.
- Repeat measurements and calculate a mean when random variation is present.
- To measure a small thickness, measure a stack of many identical objects and divide by their number.
- Quote precision that matches the measuring instrument; do not invent extra digits.

Worked example: thickness

A stack of 40 identical cards is 18.0 mm thick. Thickness of one card = $18.0 \div 40 = 0.450$ mm. Measuring one card directly would give a much larger percentage uncertainty.

Examiner insight

The June 2026 report noted that some candidates forgot to keep a stack straight, place the ruler vertically, or divide the total thickness by the number of discs. A complete practical answer states apparatus, method, repeats and calculation.

Quick check

Classify each as scalar or vector: speed, velocity, distance, displacement, mass, weight. Then explain why speed and velocity can have the same numerical value but different meanings.

2. Describing motion

Distance is total path length; displacement is change in position in a stated direction. Speed is distance travelled per unit time. Velocity is speed in a given direction. Acceleration is change in velocity per unit time.

$$v = s/t \quad \text{average speed} = \text{total distance/total time} \quad a = (v - u)/t$$

Graphs

Graph	Gradient means	Area under graph means
distance-time	speed	not used in this syllabus
speed-time	acceleration	distance travelled

Graph language: horizontal on a distance-time graph means stationary; horizontal on a speed-time graph means constant non-zero speed. A curved speed-time graph represents changing acceleration.

Worked example: speed-time area

A cyclist accelerates uniformly from 0 to 8.0 m/s in 4.0 s, then travels at 8.0 m/s for 6.0 s.
Distance = triangular area + rectangular area = $\frac{1}{2} \times 4.0 \times 8.0 + 6.0 \times 8.0 = 16 + 48 = 64$ m.

Examiner insight

A recurring error in the supplied 2026 report was confusing gradient with area: candidates calculated acceleration when distance was required, or used a rectangle where the graph formed a triangle. Sketch the shape and label base and height before calculating.

3. Mass, weight and density

Mass measures the quantity of matter and is measured in kilograms. It does not change when location changes. Weight is the gravitational force acting on a mass and is measured in newtons. Near Earth, gravitational field strength g is approximately 9.8 N/kg.

$$W = mg \quad g = W/m \quad \rho = m/V$$

Density is mass per unit volume. A regular solid can be measured geometrically; an irregular solid that sinks can be immersed in a measuring cylinder and its volume found from the rise in liquid level. Keep units consistent: 1 cm³ = 1 mL, but converting between g/cm³ and kg/m³ requires care.

Worked example: density

A metal block has mass 540 g and volume 200 cm³. $\rho = m/V = 540/200 = 2.70$ g/cm³. Because this is greater than the density of water (about 1.0 g/cm³), the unsupported block sinks.

Worked example: weight

A 6.0 kg object near Earth has weight $W = mg = 6.0 \times 9.8 = 58.8$ N. On a planet where $g = 3.7$ N/kg, its mass is still 6.0 kg but its weight is 22.2 N.

Common misconception

Do not describe g as weight or mass. It is force per unit mass. Do not write density = mass $\times$ volume; use the equation triangle or rearrange $\rho = m/V$ carefully.

4. Forces and momentum

A force can change an object's speed, direction or shape. The resultant force is the vector sum of all forces. If the resultant is zero, an object remains at rest or continues at constant velocity. Constant speed does not mean "no forces"; it means the forces balance.

$$F = ma \quad p = mv \quad \text{impulse} = F\Delta t = \Delta p$$

Friction and drag oppose relative motion and can transfer energy thermally. When a falling object reaches terminal speed, upward drag equals downward weight, so resultant force and acceleration are zero.

Momentum is mass $\times$ velocity and therefore has direction. In an isolated system, total momentum before an interaction equals total momentum after it. Momentum conservation is especially useful for collisions and explosions.

Worked example: acceleration

A 1200 kg car experiences 3000 N forward driving force and 600 N resistive force. Resultant force = 2400 N. $a = F/m = 2400/1200 = 2.0 \text{ m/s}^2$ forward.

Worked example: momentum

A 0.40 kg trolley moving at 3.0 m/s sticks to a stationary 0.20 kg trolley. Initial momentum = $0.40 \times 3.0 = 1.2 \text{ kg m/s}$. Combined mass = 0.60 kg, so final speed = $1.2/0.60 = 2.0 \text{ m/s}$.

5. Springs, moments and stability

For an elastic spring below its limit of proportionality, extension is directly proportional to force: $F = kx$. The spring constant k measures stiffness. A straight line through the origin on a force-extension graph shows proportional behaviour.

The moment of a force describes its turning effect about a pivot:

$$\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$$

For equilibrium, resultant force = 0 and total clockwise moment = total anticlockwise moment. The distance must be perpendicular to the line of action of the force.

Centre of gravity is the point through which the entire weight of an object may be considered to act. An object is more stable when it has a low centre of gravity and a wide base. It topples when the vertical line through its centre of gravity falls outside its base.

Worked example: moments

A 180 N load acts 0.40 m from a pivot. It is balanced by force F acting 0.60 m on the other side. $180 \times 0.40 = F \times 0.60$, so $F = 72/0.60 = 120 \text{ N}$.

Examiner insight

The supplied report identifies two recurring errors: rearranging the moment equation incorrectly and assuming the forces on each side must be equal. Balance moments, not necessarily forces; different distances allow different forces.

6. Energy, work, power and efficiency

Energy is stored as kinetic, gravitational potential, elastic, chemical, nuclear, electrostatic or internal energy. Energy is transferred mechanically, electrically, by heating or by waves. Total energy is conserved, although some becomes less useful and is dissipated to the surroundings.

$$E_k = \frac{1}{2}mv^2 \quad \Delta E_p = mg\Delta h \quad W = Fd \quad P = W/t = \Delta E/t$$

Work done equals energy transferred when a force moves its point of application through a distance in the force direction. Power is the rate of doing work or transferring energy. Efficiency compares useful output with total input and may be written as a decimal or percentage.

$$\text{efficiency} = \text{useful energy output} / \text{total energy input} \times 100\%$$

Worked example: energy chain

A 2.0 kg bag is lifted vertically through 3.0 m in 4.0 s. Increase in GPE = $2.0 \times 9.8 \times 3.0 = 58.8$ J. Minimum average power = $58.8/4.0 = 14.7$ W.

Worked example: efficiency

A motor receives 500 J and transfers 360 J usefully. Efficiency = $360/500 \times 100\% = 72\%$. The remaining 140 J is dissipated, mainly to thermal energy stores.

Energy resources

Compare resources using renewability, availability, reliability, scale, environmental impact and cost. Solar radiation ultimately drives wind, waves, hydroelectricity and biofuels; geothermal, nuclear and tidal energy are exceptions.

7. Pressure

Pressure is force per unit area. The same force produces greater pressure when it acts over a smaller area. Always check the area unit before calculating.

$$p = F/A$$

Pressure in a liquid increases with depth and density because a deeper point supports a taller column of liquid. In a gas, pressure results from particle collisions with container walls; at constant temperature, decreasing volume makes collisions more frequent and increases pressure.

Worked example: solid pressure

A box exerts a normal force of 640 N over an area of 0.80 m². $p = 640/0.80 = 800$ Pa. If it rests on half the area, pressure doubles to 1600 Pa.

Examiner insight

A common error in the supplied 2026 report was using pressure = force $\times$ area. Use meaning as a check: spreading a force over a larger area must reduce pressure, so area belongs in the denominator.

Equation selection checklist

- Write the quantity required and its unit.
- List known quantities with units; convert first.
- Choose an equation containing the required and known quantities.
- Rearrange symbolically, substitute, calculate and check reasonableness.

Exam technique from the 2026 evidence

The supplied question papers and examiner report show that success depends on more than recalling a formula. The same ideas are assessed through calculations, graphs, explanations and unfamiliar applications.

High-value habits

1. Show every substitution and rearrangement. 2. Distinguish area from gradient on graphs. 3. Name both the energy store and the object/surroundings. 4. For constant velocity, state that forces are balanced and resultant force is zero. 5. In practical questions, state what is measured, how repeats are used, and how the final value is calculated.

Command words

state	Give a concise fact; explanation is not required.
describe	Give relevant features, trends or steps.
explain	Link cause and mechanism to outcome.
determine	Use the information to obtain an answer, usually showing working.
show that	Present working that reaches the given result.
suggest	Apply knowledge to an unfamiliar context; more than one answer may be valid.

Five-minute retrieval

- Write seven equations from memory and add units.
- Explain the difference between mass and weight.
- State what gradient and area represent on both motion graphs.
- Explain why a low centre of gravity increases stability.
- Describe energy transfers in one everyday device.

Practice Paper

Topic P1

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Instructions

Answer all questions. For multiple-choice questions, select one option. For calculations, show working in the open space provided. Write explanations on the lines. Use $g = 9.8 \text{ N/kg}$ unless another value is given.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes. A separate worked solution section follows the answer booklet.

Section A · Multiple choice

Choose the best answer for each question.

1. Which quantity is a vector? [1]
- A distance
 - B speed
 - C energy
 - D momentum
2. What does the area under a speed-time graph represent? [1]
- A acceleration
 - B distance travelled
 - C resultant force
 - D average power
3. Which is a correct unit for density? [1]
- A kg/m^3
 - B N/kg
 - C N/m^2
 - D kg m/s
4. An astronaut travels from Earth to the Moon. Which statement is correct? [1]
- A mass and weight both decrease
 - B mass decreases but weight is unchanged
 - C mass is unchanged but weight decreases
 - D mass and weight are unchanged
5. A car moves in a straight line at constant velocity. What is its resultant force? [1]
- A zero
 - B equal to its weight
 - C equal to its driving force
 - D greater than its resistance
6. The moment of a force about a pivot is force multiplied by [1]
- A distance travelled
 - B any distance from the pivot
 - C perpendicular distance from the pivot
 - D time for which the force acts

7. Which energy store increases when a stationary object begins to move? [1]

- A chemical
- B kinetic
- C nuclear
- D gravitational potential

8. Which equation defines power? [1]

- A $P = E \times t$
- B $P = E/t$
- C $P = F/A$
- D $P = mv$

9. The same force acts over half the original area. The pressure [1]

- A halves
- B stays the same
- C doubles
- D becomes four times larger

10. A spring obeys Hooke's law. If the force doubles, the extension [1]

- A halves
- B stays the same
- C doubles
- D becomes four times larger

Section B · Structured questions

11 A runner increases velocity uniformly from 2.0 m/s to 8.0 m/s in 3.0 s.

(a) Calculate the acceleration of the runner. [2]

(b) Calculate the distance travelled by the runner during the 3.0 s interval. [2]

(c) State what the gradient of a speed-time graph represents. [1]

(d) A train travels 1.8 km in 90 s and then 3.2 km in 160 s. Calculate its average speed for the complete journey. [3]

[Total: 8]

12 A student investigates a rectangular metal block of dimensions 5.0 cm × 4.0 cm × 3.0 cm. The mass of the block is 162 g.

(a) Calculate the volume of the block. [1]

(b) Calculate the density of the block. [2]

(c) The block is taken from Earth to the Moon. State what happens to (i) its mass and (ii) its weight. [2]

(d) Describe how to determine the density of an irregular stone that sinks in water. Include the apparatus, measurements and calculation. [3]

[Total: 8]

13 A car of mass 900 kg has a forward driving force of 2500 N and a total resistive force of 700 N.

(a) Calculate the resultant force on the car. [1]

(b) Calculate the acceleration of the car. [2]

(c) The car later travels at constant velocity. State the resultant force and explain your answer. [2]

(d) A 0.25 kg ball moving at 12 m/s rebounds in the opposite direction at 8.0 m/s. Calculate the magnitude of the change in momentum of the ball. [3]

[Total: 8]

14 A child of weight 240 N sits 1.5 m from the pivot of a seesaw. An adult of weight 360 N sits on the opposite side.

(a) Calculate the moment of the child's weight about the pivot. [2]

(b) Calculate the distance from the pivot at which the adult must sit to balance the seesaw. [2]

(c) State one condition, other than balanced moments, required for an object to be in equilibrium. [1]

(d) Explain why a vehicle with a low centre of gravity and a wide base is more stable. [3]

[Total: 8]

15 A crane lifts a 250 kg load vertically through 8.0 m in 20 s. The input power to the crane is 12.0 kW.

(a) Calculate the increase in gravitational potential energy of the load. [2]

(b) Calculate the useful power output of the crane. [2]

(c) Calculate the efficiency of the crane.

[2]

(d) A person of weight 620 N stands on shoes with a total contact area of 0.040 m².
Calculate the pressure exerted on the floor.

[2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper. Mark only after finishing the full attempt.

Marking routine

Award marks for correct physics and method, not only the final number. Circle the first error in each response and classify it as: knowledge, equation choice, rearrangement, unit conversion, graph interpretation or communication.

Solutions · Multiple choice

1. 1 D - momentum has magnitude and direction.
2. 2 B - area under a speed-time graph equals distance travelled.
3. 3 A - density may be measured in kg/m^3 .
4. 4 C - mass is unchanged; weight is smaller because gravitational field strength is smaller.
5. 5 A - constant velocity means zero acceleration and therefore zero resultant force.
6. 6 C - the distance is perpendicular to the force's line of action.
7. 7 B - kinetic energy increases with speed.
8. 8 B - power is energy transferred per unit time.
9. 9 C - $p = F/A$, so halving area doubles pressure.
10. 10 C - below the limit of proportionality, $F = kx$.

Solutions · Structured questions

11(a) [2] $a = (v - u)/t = (8.0 - 2.0)/3.0 = 2.0 \text{ m/s}^2$. Award 1 mark for substitution and 1 for the answer with unit.

11(b) [2] distance = average speed $\times$ time = $(2.0 + 8.0)/2 \times 3.0 = 15 \text{ m}$. Award 1 mark for the valid method and 1 for the answer.

11(c) [1] Acceleration.

11(d) [3] Total distance = $1800 + 3200 = 5000 \text{ m}$; total time = 250 s ; average speed = $5000/250 = 20 \text{ m/s}$. Award 1 mark for each stage.

12(a) [1] $V = 5.0 \times 4.0 \times 3.0 = 60 \text{ cm}^3$.

12(b) [2] $\rho = m/V = 162/60 = 2.7 \text{ g/cm}^3$. Award 1 mark for the equation/substitution and 1 for the answer.

12(c) [2] Mass remains unchanged [1]. Weight decreases [1] because the Moon has a smaller gravitational field strength.

12(d) [3] Measure mass with a balance [1]. Find volume from the rise in water level in a measuring cylinder when the stone is fully immersed [1]. Calculate density = mass/volume [1].

13(a) [1] Resultant force = $2500 - 700 = 1800 \text{ N}$ forward.

13(b) [2] $a = F/m = 1800/900 = 2.0 \text{ m/s}^2$. Award 1 mark for substitution and 1 for the answer.

13(c) [2] The resultant force is zero [1] because constant velocity means zero acceleration / the driving and resistive forces are balanced [1].

13(d) [3] Taking the initial direction as positive: initial momentum = $0.25 \times 12 = 3.0 \text{ kg m/s}$ [1]; final momentum = $0.25 \times (-8.0) = -2.0 \text{ kg m/s}$ [1]; magnitude of change = 5.0 kg m/s [1].

14(a) [2] moment = force $\times$ perpendicular distance = $240 \times 1.5 = 360 \text{ N m}$. Award 1 mark for the method and 1 for the answer.

14(b) [2] $360 \times d = 360 \text{ N m}$, so $d = 360/360 = 1.0 \text{ m}$. Award 1 mark for the moment equation and 1 for the answer.

14(c) [1] Resultant force equals zero.

14(d) [3] A low centre of gravity keeps its line of action lower [1]; a wide base makes it harder for that line to pass outside the base [1]; therefore a larger turning effect / tilt is required before toppling [1].

15(a) [2] $\Delta E_p = mg\Delta h = 250 \times 9.8 \times 8.0 = 19\,600 \text{ J}$. Award 1 mark for substitution and 1 for the answer.

15(b) [2] $P = E/t = 19\,600/20 = 980 \text{ W}$. Award 1 mark for the method and 1 for the answer.

15(c) [2] efficiency = $980/12\,000 \times 100\% = 8.17\%$ (accept 8.2%). Award 1 mark for substitution and 1 for the answer.

15(d) [2] $p = F/A = 620/0.040 = 15\,500 \text{ Pa}$. Award 1 mark for substitution and 1 for the answer.

Self-assessment

Mark each answer, circle the first incorrect step, and label the cause: knowledge, equation choice, rearrangement, unit conversion, graph interpretation or communication. Your label determines what to revise next.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.